# Modeling Post-Wildfire Hydrologic Regime Change Using a Data-Driven SWAT+ Calibration Framework

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Submitted to *Hydrological Processes*

Keywords: SWAT+, Wildfire, Hydrological responses, Post-wildfire hydrology, Water balance alteration, Runoff processes

## Abstract

Understanding post-wildfire hydrologic change is essential for reliable water-resources forecasting and post-fire management. This study uses the semi-distributed SWAT+ model to quantify the effects of the August Complex Fire (2020) on the Middle Fork Eel watershed in northwestern California by comparing a "with-fire" simulation to a counterfactual "without-fire" simulation over the same post-fire period (November 2020-December 2023). Model performance was first established for pre-fire conditions using USGS streamflow. On the daily scale, calibration and validation produced Nash-Sutcliffe Efficiency (NSE) / Kling-Gupta Efficiency (KGE) values of 0.67/0.82 and 0.76/0.72 respectively and on the monthly scale, NSE/KGE were 0.88/0.90 (calibration) and 0.92/0.83 (validation). For the post-fire period, a data-driven bidirectional calibration clearly outperformed a literature-constrained, one-directional setup and was adopted as the best model, achieving daily NSE/KGE of 0.68/0.84 and 0.69/0.83 for calibration and validation respectively. Using this best-performing configuration, results show that wildfire drove a persistent shift from storage-dominated to a quick-runoff regime. During wet seasons, surface runoff and lateral flow increased sharply, while deep percolation declined by 50-200 mm per month, indicating strong reduction of vertical infiltration and groundwater recharge. Evapotranspiration (ET) decreased in dry months probably because of canopy loss but increased in wet months possibly due to enhanced bare-soil evaporation, leading to net annual ET gains. Streamflow peaks became larger and more frequent signaling loss of watershed buffering capacity due to the wildfire impact. Overall, the fire created a flashy but dry system: more water is exported during storms, but long-term groundwater replenishment is reduced, highlighting the need for post-fire treatments that restore vertical infiltration pathways and protect downstream water security.


## 1. Introduction

Wildfire is a critical ecological process and growing global management concern, with increasing frequency and severity in the U. S. and elsewhere (Doerr and Santín, 2016; Havel et al., 2018; Wampler et al., 2023). These events significantly alter terrestrial ecosystems and hydrologic responses within affected watersheds (Shakesby and Doerr, 2006). Assessing wildfire impacts on hydrology and water quality is increasingly important due to rising wildfire trends and the dependence on water from fire-prone regions (Ebel et al., 2023; Robinne et al., 2021). Post-fire conditions often result in sharply increased runoff and erosion (Pierson et al., 2001; Benavides-Solorio and MacDonald, 2001; Ryan et al., 2011; Inbar et al., 1998; Wine and Cadol, 2016), heightening flood and debris flow risks in vulnerable, often populated areas (Neary et al., 2003). Wildfires affect both the timing and volume of surface and subsurface flow, reducing water supply reliability (Bladon et al., 2014; Hallema et al., 2017; Hallema et al., 2018; Williams et al., 2022), while affecting water quality (Murphy et al., 2018; Smith et al., 2011), carbon storage, and ecosystem stability. Changes in infiltration, interception, and hydraulic roughness lead to shifts in peak flows, and reduced transpiration post-fire can increase annual water yield. Wildfires drive major changes in land use-land cover and biomes, influencing hydrologic regimes by reducing canopy interception and evapotranspiration (ET), while increasing runoff, velocity, and reducing infiltration and storage due to loss of ground cover. These impacts alter

key hydrologic components like surface runoff, streamflow, ET, subsurface flow, and total water yield (Coombs et al., 2013; Venkatesh et al., 2020). The extent of these changes is typically greater in high burn intensity areas depending on types and density of vegetation, soil texture, and topography.

Numerous studies have investigated the hydrological effects of wildfire across various ecosystems, including forests and rangelands (Coombs and Melack, 2013; Liu et al., 2004; Mahat et al., 2016). These investigations frequently document a marked increase in runoff and erosion rates immediately after a fire (Shakesby and Doerr, 2006). Increased peak discharge and stormflow volumes are also commonly reported findings. Changes in infiltration rates are a key driver of altered runoff, with studies highlighting the role of fire-induced soil water repellency in impeding percolation and promoting overland flow (Martin and Moody, 2001; Moody and Martin, 2001; Robichaud, 2000; Nyman et al., 2010). Alterations to subsurface flow and base flow have also been observed generally showing a decrease under post-wildfire conditions as surface runoff increases. However, some studies suggest increased baseflow through decreased ET or increased flow magnitudes related to groundwater recharge. Other studies have shown that ET is typically reduced after a wildfire event due to vegetation loss, contributing to increased total water yield (Maina and Siirila-Woodburn, 2020; Nasirzadehdizaji and Akyuz, 2022; Townsend and Douglas, 2004). For example, a study of coastal California chaparral watersheds after a 2004 wildfire and subsequent heavy rainfall aimed to determine changes in hydrology and the export of nutrients and suspended sediment, finding significantly higher peak discharge and sediment export from burned areas (Coombs et al., 2013). Studies also indicate that moderate to high severity burns particularly impact infiltration rates, leading to increased surface runoff (Ebel et al., 2020; Scott et al., 1990; Shakesby and Doerr, 2006), and evergreen forests, as a dominant canopy type, are susceptible to changes like decreased ET following fire due to vegetation mortality (Maina and Siirila-Woodburn, 2020; Baur et al., 2024). J.A.H. Brown. (1972) conducted research in south-eastern New South Wales, Australia, after a severe 1965 bushfire and sought to understand how the fire altered flood hydrograph shape, peak discharge, catchment runoff, and sediment loads, demonstrating pronounced changes and significantly larger sediment loads in the years following the fire. Mahat et al. (2016) conducted an investigation in Alberta, Canada, following the 2003 Lost Creek wildfire examined longer-term hydrological effects, reporting higher mean annual water yield and peak flows in burned catchments, though noting the confounding influence of precipitation variability. Such research was designed to characterize and quantify hydrologic responses across various spatial scales, from high burn intensity sub-watersheds to the entire watershed scale.

Process-based hydrologic modelling is widely used to understand the complex, variable conditions following wildfires, particularly in watershed processes (Ebel et al., 2023; Kiesel et al., 2013; Moody et al., 2013; Beven, 2001; Famiglietti and Wood, 1995). Semi-distributed physically based models such as SWAT are especially useful for simulating natural and human-induced disturbances by representing multiple watershed processes (Arnold et al., 1998; Clark et

al., 2017; Neitsch et al., 2011). Wildfire modelling typically involves adjusting land use/land cover (LULC) and parameters such as the runoff curve number (CN) to reflect post-fire conditions (Venkatesh et al., 2020; Batelis and Nalbantis, 2014). This includes modifying land use in burned areas or using fire susceptibility maps to simulate post-fire scenarios and update SCS-CN and soil erodibility. SWAT model as well as the next-generation version of the original SWAT model, SWAT+, has been successfully used for wildfire impact studies, due to its integration with satellite-based data tools (McDonald et al., 2019), robust sediment transport algorithms and scalability for long-term hydrological simulations (Nkwasa et al., 2022). Its flexibility in calibration (Niroula et al.,2023) and compatibility with hybrid modeling (Esmaeilzadeh and Amirzadeh, 2024) further strengthens its utility. Overall, while alternative physical and non-physical models excel in specific domains, SWAT+ offers a comprehensive and adaptable framework that continues to make it a strong choice for large-scale and disturbance-focused watershed analyses.

This study investigates how wildfire alters watershed hydrology by explicitly comparing two SWAT+ simulations: a “with fire” scenario representing post-wildfire conditions and a counterfactual “without fire” scenario representing the hypothetical hydrologic response over the same period in the absence of burning. The objective is to quantify and analyze the impact of wildfire on key hydrologic processes-surface runoff, infiltration, evapotranspiration, lateral flow and streamflow, addressing the central research question: How do the wildfire modify the hydrologic regime of the watershed relative to the response that would have occurred without fire? To represent these fire effects, the study employs the Land Use Update functionality in SWAT+ to impose burn-severity-dependent changes, enabled by compiling and implementing the modified source code provided by Arnold et al. (2024), as this feature is inactive in the current public release. Also, this study investigates whether a data-driven wildfire calibration framework in SWAT+, combining burn-severity-dependent land-use updates with automatic, bidirectional parameter calibration, be constrained using post-fire streamflow observations alone to exhibit physically meaningful changes in watershed parameters, using SWAT+ Toolbox (Celray James Chawanda, 2022) instead of literature based empirical adjustment.

## 2. Study Area

The August Complex wildfire (August-October 2020; CAL FIRE, 2023), which severely affected medium-sized, predominantly evergreen forest Middle Fork Eel Watershed that has been largely understudied despite its high wildfire vulnerability, is used as an experimental wildfire for this study. Located in northwestern California’s Coastal Range, the Middle Fork Eel Watershed (**Figure 1**) spans ~175 square miles and features steep, erosion-prone terrain with elevations ranging from 300 to 2,200 meters. Its Mediterranean climate brings intense winter rainfall (1,000-1,500 mm annually) and prolonged summer droughts, creating a hydrologic regime dominated by flashy winter flows and low summer baseflows (Jansen et al., 2020). Its rugged terrain and steep elevation gradients significantly influence snowmelt timing and sediment transport, making the watershed especially sensitive to post-fire hydrologic changes (Scott and

Wohl, 2018). Ecologically, the watershed supported diverse aquatic life before fire disturbances (Saunders et al., 2018). The August Complex Fire, California's largest recorded wildfire at the time, burned over 1 million acres, including ~65% of the Middle Fork Eel Watershed in 2020 (CAL FIRE, 2023), leading to major changes in land cover and altering the area's hydrological dynamics. These factors make the Middle Fork Eel Watershed an important location for studying how wildfires influence watershed hydrology, particularly in terms of land cover transformation and changes in water flow and storage.

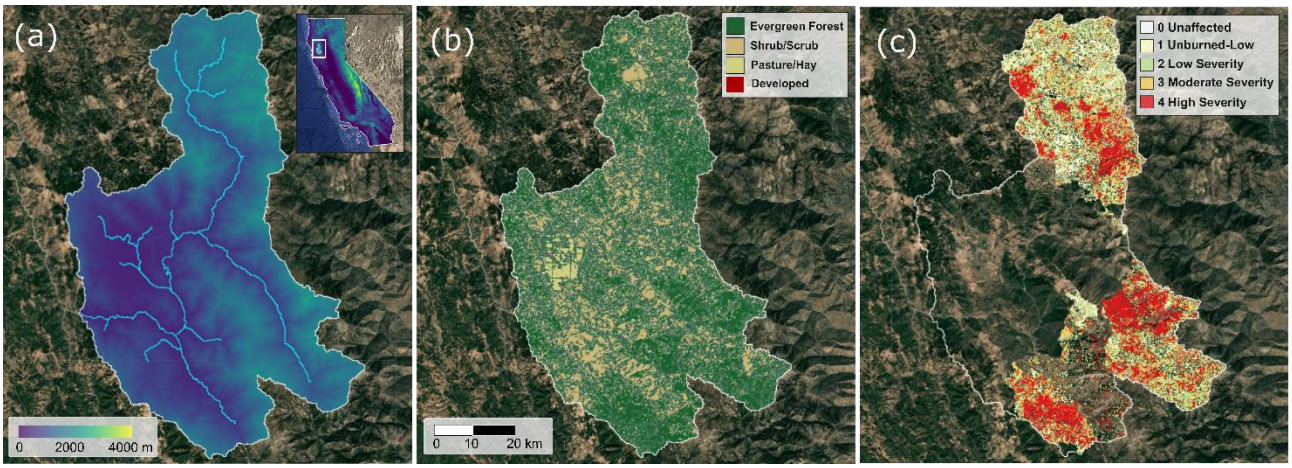


*Figure 1. Middle Eel Fork Watershed, California, along with (a) digital elevation model (DEM), (b) land use/land cover map, and (c) Burn Severity map overlaid with actual HRUs.*

# 3. Methodology

## 3.1 SWAT+ Hydrological Modeling

The SWAT is a continuous, distributed-parameter, process-based watershed model used for assessing the impact of land management on water resources and nonpoint source pollution (Arnold et al., 1998; Neitsch et al., 2011; Havel et al., 2018). It simulates runoff, infiltration, and ET by dividing a watershed into sub-basins, which are further split into Hydrologic Response Units (HRUs) defined by unique land use, soil, and slope combinations. Surface runoff is commonly simulated using SCS-CN method (Batelis & Nalbantis, 2014; Venkatesh et al., 2020; Nasirzadehdizaji & Akyuz, 2022; Ebel et al., 2023), which utilizes the fixed relationship between rainfall, soil type, land use, and antecedent moisture conditions. To simulate wildfire impacts in the original SWAT, users typically modify parameters and inputs to reflect post-fire conditions. This includes increasing the CN to account for reduced infiltration, altering land cover classifications (e.g., to reflect canopy loss), and adjusting soil properties like saturated hydraulic conductivity (SOL_K) and available water capacity (SOL_AWC) (Ebel & Moody, 2020). This parameter modification can be time-consuming and computationally intensive in the original SWAT. SWAT+ is a completely restructured version of SWAT designed for greater flexibility (Bieger et al., 2017). A key change is the introduction of landscape units (LSUs), which allow for routing water flow and pollutants within sub-basins. In SWAT+, spatial objects like HRUs,

channels, and aquifers are treated as distinct entities with user-defined connections. This enhanced spatial representation and routing capability in SWAT+ offers potential for a more realistic simulation of the spatial heterogeneity caused by wildfires compared to the original model's HRU structure. SWAT+ also features a dedicated calibration file for easier parameter adjustments. As mentioned earlier, since the current SWAT+ public version ([Computer software] SWAT+ Development Team, 2025) does not have land use update decision table features active required for our study, the source code of the unreleased version capable of doing that was obtained from GitHub repository (Arnold et al., 2024) and was compiled to create the executables capable of simulating land use change.

### 3.2 Data Sources

To run the SWAT+ model, several input and ancillary datasets are required. Topographic data was obtained from the NASA Jet Propulsion Laboratory Shuttle Radar Topography Mission (SRTM) Digital Elevation dataset (USGS, 2018), available at 30-meter resolution detailed and validated by Farr et al. (2007). Land use/Landcover (LULC) was characterized using the Annual NLCD Collection 1 Science Products (USGS, 2024). with the spatial resolution of 30 m. Soil data was obtained from FAO/UNESCO Soil Map of the World (FAO/UNESCO, 1974-1981). Meteorological input variables, including 4-km gridded daily precipitation and maximum and minimum temperature, were sourced from the PRISM (Parameter-elevation Regressions on Independent Slopes Model) dataset (PRISM Group, 2014), with methodological development and validation described by Daly et al. (2008). Burn severity data were acquired from the MTBS (Monitoring Trends in Burn Severity) viewer (USGS & U.S. Forest Service, 2025) with 30 m spatial resolution, which maps the extent and severity of large fires using remote sensing satellites such as Landsat TM/ETM+/OLI or Sentinel-2 MSI (Eidenshink et al., 2007). Burn severity mapping was conducted using MTBS-derived thematic burn severity classes, which classify burned areas such as unburned to low, low, moderate, and high severity. Streamflow data was collected from USGS National Water Dashboard (USGS, 2025a) Gage station MF EEL R NR DOS RIOS 11473900, providing standardized records of river discharges.

### 3.3 Experimental Setup for Model Simulation

The model setup began with watershed delineation using a DEM. Following this, spatial datasets detailing LULC, soil type, and slope classes were integrated (Havel et al., 2018). Within each sub-watershed, HRUs were defined based on unique combinations of these spatial attributes. To focus on the dominant landscape characteristics and in accordance with standard modelling practices (Neitsch et al., 2011), a threshold of 5% area was applied during HRU definition to eliminate minor combinations. This process identified three dominant land cover classes for the study area: Evergreen Forest, shrubland, and pastureland. SWAT+ allows flexibility in watershed configuration, treating HRUs, channels, and other elements as distinct spatial objects (Bieger et al., 2017). To assess the hydrological impacts of wildfire, two main scenarios were simulated:

(1) A baseline (Pre-Fire/without fire) Scenario, representing undisturbed hydrological conditions influenced by existing vegetation and soil properties. Calibration and validation were done using streamflow data from 2005 to 2019, later extended in the post-fire period, and the calibrated parameters with the calibration type and value are provided in **Supplementary Table A1**.

(2) A post-fire/with fire Scenario, configured to reflect the significant landscape and hydrological alterations caused by wildfire (Havel et al., 2018; Wampler et al., 2023).

The implementation of with-fire conditions involved changing the land use after the wildfire to represent the wildfire impact and then modifying model parameters to reflect the altered landscape. A burn severity map was overlaid with the HRU map to identify burned areas, as shown in Supplementary Figure B1. However, since SWAT+ analysis is done on HRUs level instead of grid/pixel level, to parameterize wildfire effects within the model, an area-weighted burn severity was first determined for each HRU in all land-use classes by overlaying the burn-severity map onto the HRU layer. This provided each HRU with a representative, area-weighted burn-severity value and number of HRUs within each severity class (**Figure 2**). By applying the decision table function, low-severity HRUs remained in their original land use but were reduced by 25% of their biomass to mimic low fire effect, while moderate-severity HRUs were changed to rangeland (RNGE), based on Wampler et al. (2023) and USGS burn severity glossary (USGS, 2025b). The decision tables used are provided in **Supplementary Tables C1 and C2**. No HRUs were identified as high severity; therefore, this class was eliminated from further examination. Then, burn severity-based parameter adjustments were then introduced into the low- (2) and moderate- (3) severity classes, following observations by Wampler et al. (2023) and Havel et al. (2018) and further adjusted and verified with post-fire discharge data (November 2020-December 2022) using SWAT+ Toolbox. Unburned (0) and unburned-to-low-severity (1) regions were left unaffected because they did not experience wildfire.

Then for model parameters adjustment to incorporate wildfire impact, to account for the high predictive uncertainty inherent in post-wildfire hydrological modeling (Ebel et al., 2023) and to address the problem of equifinality in distributed models (Beven, 2001), a dual-strategy calibration approach was employed using the DDS algorithm. We tested two distinct parameterization hypotheses to determine whether the observed post-fire streamflow was sufficiently strong to drive the model toward physical reality without imposing rigid theoretical constraints. The first approach, a Literature-Constrained (Unidirectional) calibration, strictly forced parameters to move in directions conventionally associated with fire impacts (e.g., CN2 was only allowed to increase; CANMX was only allowed to decrease). The second approach, a Data-Driven (Bidirectional) calibration, allowed parameters to vary bidirectionally within symmetric ranges (e.g., ±25% for Low Severity, ±50% for Moderate Severity). This served as a diagnostic test: if the algorithm independently converged toward physically realistic values without being forced, it would rigorously validate the strength of the wildfire signal in the

observed data. Parameters, their adjustment range, and final adjusted values are shown in the **Supplementary Table A2**.

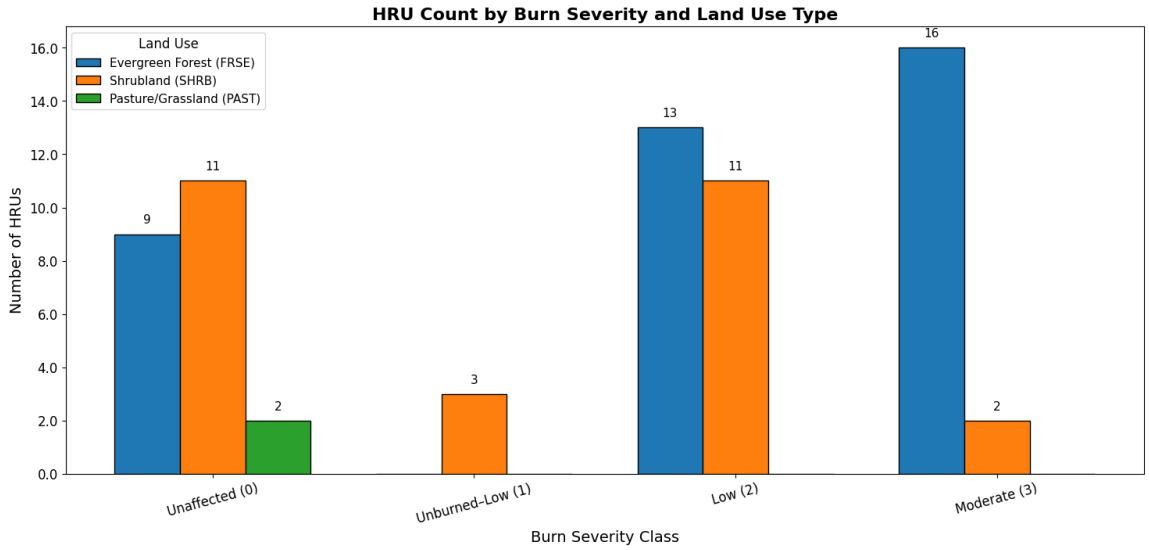


*Figure 2. Number of HRUs within each burn severity class for different land-use types. Burn severity classes definitions are as follows: 0 – unaffected, 1 – unburned to low, 2 – low, 3 – moderate, and 4 – high severity.*

### 3.4 Spectral and Statistical Methods

To evaluate the significance of the observed differences in hydrological response, both spectral and statistical techniques were employed. Wavelet analysis, described as a timescale or frequency-resolved analytical method (Torrence & Compo, 1998), was used to characterize the temporal variability within the streamflow signals (Gaucherel, 2002). This technique decomposes a signal into scaled and translated versions of a base function, allowing for the identification of patterns and energy distribution across different temporal scales (Schaefli et al., 2007). Wavelet transformation has been applied to runoff data in hydrology to analyze how water flow patterns change over time as well as to better understand the timing and frequency of runoff events (Labat et al., 2000). In this analysis, the Level 4 approximation coefficients from the discrete wavelet transform, which captures the broader, long-term trends in streamflow signals while filtering out low-frequency components, were specifically examined, and the analysis was performed using the 'wavelets' package in R. For further details about wavelet analysis in R, please review Aldrich (2020).

Non-parametric statistical tests, particularly the Wilcoxon rank-sum test and its variants, have been effectively employed in hydrological studies to assess the impacts of wildfire (Loáiciga et al., 2001; Baur et al., 2024). The Wilcoxon Rank Sum Test is a non-parametric statistical test that compares two independent groups by ranking all observations together and assessing whether the sum of ranks for one group significantly differs from that of the other, thereby estimating if the groups come from the same distribution (Wilcoxon, 1945). For instance, Loáiciga et al. (2001) applied the Wilcoxon test to determine statistically significant differences in mean streamflow

between fire and non-fire years in a chaparral watershed, using long-term observational data. More recently, Baur et al. (2024) employed a two-sided Wilcoxon rank-sum test to compare pre- and post-fire distributions of soil moisture and vegetation water content change rates, further demonstrating its applicability to post-fire ecohydrological responses. Building on this precedent, the Wilcoxon rank-sum test is used in this study to statistically assess whether wildfire has a significant effect on streamflow by comparing distributions from "with-fire" and "without-fire" scenarios during the post-wildfire period. The null hypothesis is that there is no significant change in streamflow due to wildfire, and if the Wilcoxon test statistic value $V > 0$ and $p < 0.05$ at 5% significance level, the null hypothesis is rejected, i.e., Wildfire has caused a significant alteration in the streamflow. The Wilcoxon rank-sum test was performed using the base R package. Learn more at R Documentation Contributors (2023).

However, wildfire-induced discharge data is often characterized by heteroscedasticity (unequal variances) and extreme skewness during storm events, which can violate the symmetrical assumptions of the Wilcoxon test. To address these limitations, we also employ the Brunner-Munzel test. The Brunner-Munzel test, well suited to environmental modeling, does not assume symmetric distributions or equal variances between the “with-fire” and “without-fire” scenarios (Brunner & Munzel, 2000). It evaluates the null hypothesis of stochastic equality. In this study, the test applied paired differences in daily discharge during the post-fire period, with the null hypothesis stating that wildfire simulation does not cause a significant shift in the streamflow distribution. This hypothesis is rejected when the p-value is less than the 5% significance level. The analysis was implemented using the brunnermunzel package in R (Toshiaki, 2022).

The Nash–Sutcliffe Efficiency (NSE), a normalized form of the Mean Squared Error (MSE), was used for assessing and calibrating the model using observed data. Because NSE is dimensionless and scaled between negative infinity and one, it provides a convenient basis for comparing model performance across different datasets and basins.

$$NSE = 1 - \frac{\sum_{t=1}^{n}\left(x_{s,t} - x_{o,t}\right)^2}{\sum_{t=1}^{n}\left(x_{s,t} - \mu_0\right)^2}$$

where n is the total number of time-steps, $x_{s,t}$ is the simulated value at time-step t, $x_{o,t}$ is the observed value at time-step t, and $\mu_0$ is the mean of the observed values (Gupta et al., 2009).

The Kling–Gupta Efficiency (KGE) was also used to evaluate the model performance by measuring how closely simulated streamflow aligns with observed data. It integrates correlation, variability, and bias components to offer a more balanced and detailed assessment of model behavior than traditional metrics such as the NSE (Knoben et al., 2019).

$$KGE = 1 - \sqrt{(r-1)^2 + (\frac{\sigma_{sim}}{\sigma_{obs}} - 1)^2 + (\frac{\mu_{sim}}{\mu_{obs}} - 1)^2}$$

where $\sigma_{obs}$ is the standard deviation in observations, $\sigma_{sim}$ the standard deviation in simulations, μsim the simulation mean, and $\mu_{obs}$ the observation mean.

Model uncertainty was quantified using the 95% prediction uncertainty (95PPU) band derived from the Dynamically Dimensioned Search (DDS) algorithm. The quality of this uncertainty quantification was assessed using the P-factor and R-factor (Abbaspour et al., 2004; Abbaspour et al., 2018). The P-factor represents the percentage of observed data that falls within the 95PPU band (defined by the 2.5th and 97.5th percentiles of the cumulative simulation distribution). Ideally, the P-factor should bracket most observations. It is computed as:

$$\text{P-factor} = \frac{1}{n}\sum_{t=1}^{n} S_t \,, S_t = \begin{cases} 1, & \text{if } y_{t,L} \leq y_{t,obs} \leq y_{t,U} \\ 0, & \text{otherwise} \end{cases}$$

The R-factor quantifies the average thickness of the uncertainty band relative to the standard deviation of the observed data and reflects the precision of the calibration. A lower R-factor (typically < 1) indicates a narrower and more constrained uncertainty band. It is calculated as:

$$\text{R-factor} = \frac{\bar{d}_{95\text{PPU}}}{\sigma_{\text{obs}}} = \frac{\frac{1}{n}\sum_{t=1}^{n}(y_{t,U} - y_{t,L})}{\sigma_{\text{obs}}}$$

Where, $y_{t,obs}$is the observed streamflow at time $t$, $y_{t,L}$and $y_{t,U}$are the lower (2.5%) and upper (97.5%) limits of the 95PPU band at time $t$, $n$ is the total number of observations; and $\sigma_{obs}$is the standard deviation of the observed streamflow.

# 4. Results

A comparative analysis of "with fire" and "without fire" SWAT+ simulations for the Middle Fork Eel Watershed reveals significant seasonal shifts in hydrological processes following the August Complex Wildfire (2020). The watershed, located in a Mediterranean climate, experiences distinct wet (Nov-Apr) and dry (May-Oct) seasons, which shape the post-fire responses across all hydrologic components.

## 4.1 SWAT+ Model Performance

For daily streamflow, the model reproduced the observed hydrograph with NSE/KGE values of 0.67/0.82 during calibration and 0.76/0.72 during validation, with corresponding P- and R-factors of 75%/0.80 and 77.3%/0.60, respectively as shown in **Figure 3**. On the monthly time step, model performance improved further, with NSE/KGE values of 0.88/0.90 for calibration and 0.92/0.83 for validation with corresponding P- and R-factors of 73.1%/1.13 and 83.3%/0.88, respectively (**Supplementary Figure A1**).

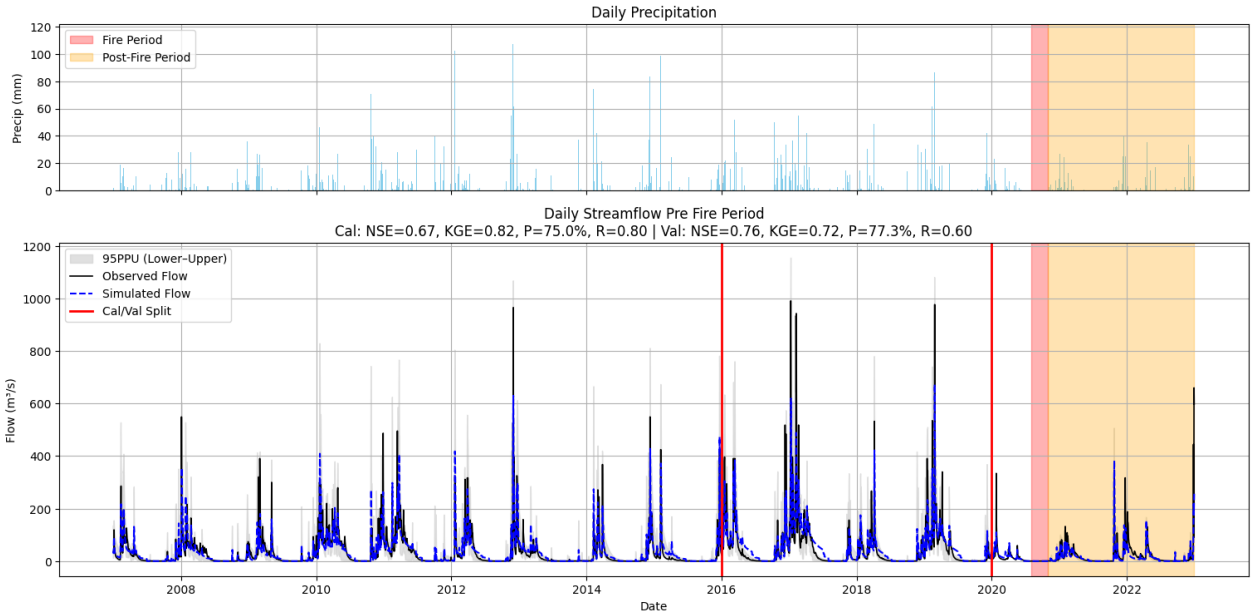


*Figure 3. Daily time series of SWAT+-simulated and USGS-observed streamflow, along with precipitation from PRISM, during the pre-fire, fire, and post-fire periods. The plot shows the calibration (2005–2016) and validation (2016–2019) windows and summarizes model performance indices (NSE, KGE, and 95PPU P- and R-factors) for each period.*

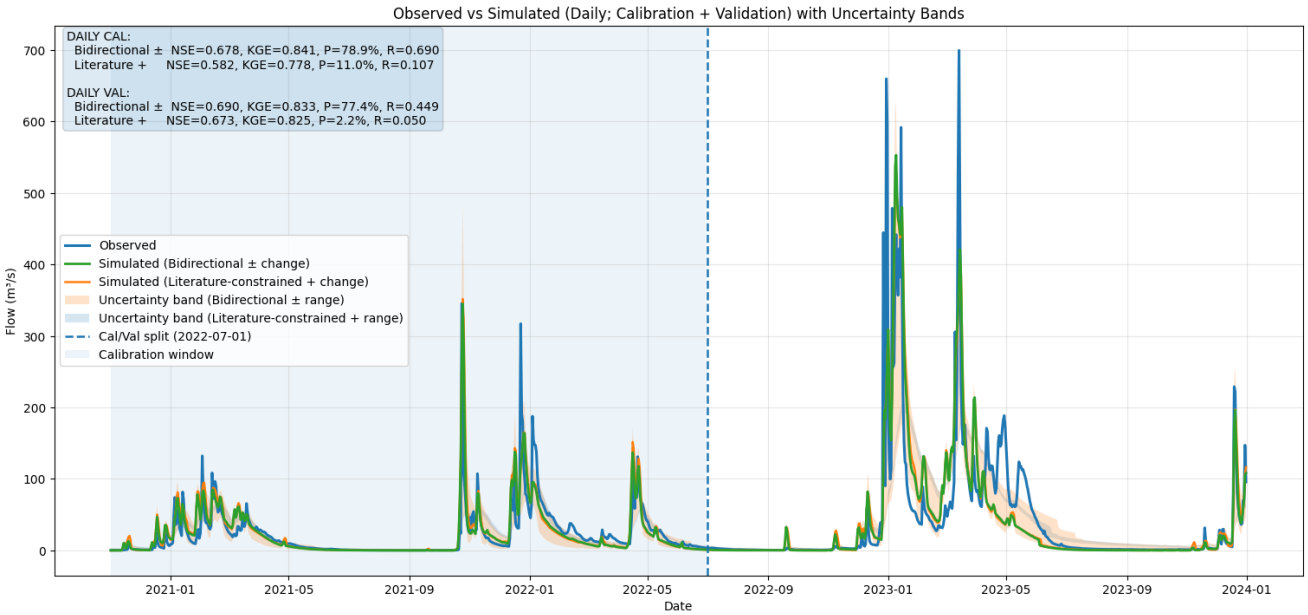


*Figure 4. Comparison of observed daily discharge with two alternative calibration frameworks: Bidirectional ± change (green) and Literature-constrained + change (orange) to incorporate wildfire impact. Shaded envelopes indicate the uncertainty range for each framework.*

The calibration results for the post-fire period revealed a critical performance disparity between the two approaches, favoring the unconstrained bidirectional methodology as shown in **Figure 4**. While the Literature-Constrained model achieved a defensible Nash–Sutcliffe Efficiency (NSE = 0.582), it resulted in a critically low P-factor of 11.0%, indicating that the rigid application of standard theoretical constraints forced the model into a solution space that excluded nearly 90% of the observed variability suggesting that standard rules of thumb for fire modeling may oversimplify complex feedback, such as the trade-off between ash storage and runoff generation (Woods & Balfour, 2010). Conversely, the Bidirectional model achieved a significantly higher NSE/KGE of 0.68/0.84 and a P-factor of 78.9% during calibration and 0.69/0.83 and a P-factor of 77.4% during validation respectively, successfully bracketing most observed flows. Monthly streamflow metrics further reinforce this conclusion (**Supplementary Figure A2**). During the calibration window, the Bidirectional model simulated monthly flows with NSE = 0.858 and KGE = 0.910, whereas the Literature-Constrained model performed noticeably less good (NSE = 0.781, KGE = 0.825) and again failed to adequately enclose observations (P-factor = 5%). During the validation period, the Bidirectional model maintained strong performance (NSE = 0.812, KGE = 0.795) relative to the Literature-Constrained model (NSE = 0.803, KGE = 0.781). Together, these daily and monthly results demonstrate that the Bidirectional model more accurately captures post-fire watershed behavior. Following the guidelines of Abbaspour et al. (2018) for disturbed basins, it was therefore selected for subsequent analysis.

### 4.2 Wavelet and Statistical Verification of Wildfire Influence on Streamflow

This analysis compared simulated streamflow from the without-fire and with-fire scenarios during the post-fire period (after November 1, 2020) to assess whether the hydrological differences observed were statistically and dynamically significant. The wavelet-based analysis (Discrete Wavelet Transform, Level 4) of daily streamflow from SWAT+ simulations reveals that wildfire significantly altered the low-frequency, baseline component of flow. **Figure 5a** shows the Level 4 wavelet approximation coefficients, where the x-axis (Index) represents the compressed time dimension after wavelet decomposition (every index corresponds to ~16 daily time steps). The y-axis (Amplitude) indicates the magnitude of the large-scale flow component at each time index. The “With Fire” scenario exhibits consistently higher peaks in amplitude during

wet periods, suggesting that wildfire increased and prolonged high-flow conditions.

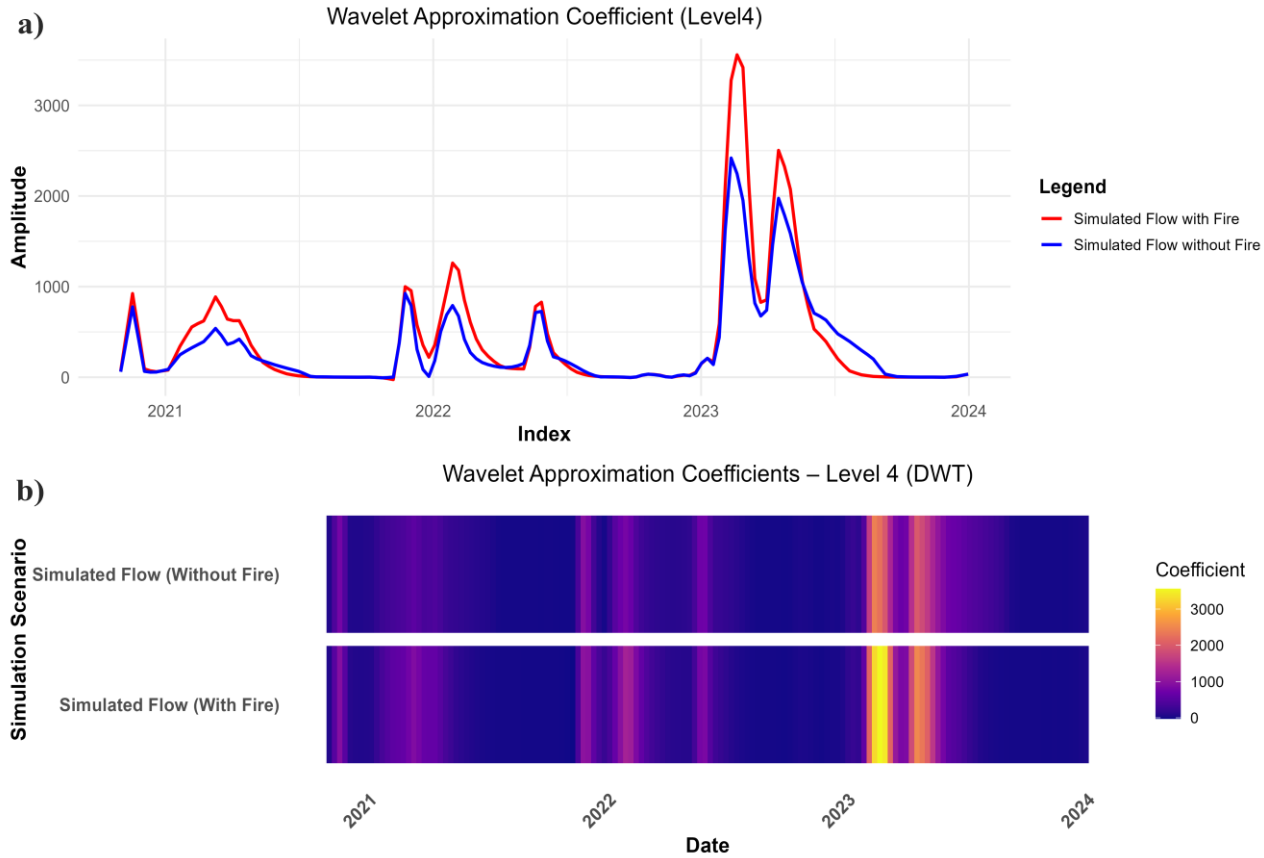


*Figure 5. Comparison of Wavelet approximation coefficients for SWAT+ simulated flow between with fire and without fire scenario (a) time series (b) heatmap*

The heatmap in **Figure 5b** reinforces this observation, with intense brightbands appearing more frequently and persistently in the "With Fire" scenario. These bands represent elevated approximation coefficients,nd indicate sustained periods of enhanced flow magnitude under post-fire conditions. Only Level 4 of the wavelet decomposition is shown here because it captures the seasonal-scale dynamics without being dominatedby short-term noise(as in Levels 1–2) or over smoothing broader trends (as in higher levels). Level 4 therefore provides an effective compromise between temporal resolution and interpretability, making it well suited to detecting how wildfire modifies longer-term hydrologic behavior and amplifies the baseline hydrograph during wet seasons.

To formally test these differences, we evaluated paired daily simulated streamflow from the "With Fire" and "Without Fire" scenarios using two complementary nonparametric approaches. The Wilcoxon signed-rank test indicated a highly significant shift in the median of the paired differences ($V = 268{,}629$, $p = 9.1 \times 10^{-9}$), confirming that daily flows in these two scenarios are not equivalent. The Brunner-Munzel test, which is robust to unequal variances and non-normal distributions, also detected a statistically significant distributional shift (BM statistic = -2.45, df = 1155, $p = 0.015$), indicating that the overall flow regime changed due to wildfire. Consistent

with the histogram of daily differences (**Figure 6a**) and the scatterplot of With-Fire versus Without-Fire flows (**Figure 6b**), these tests show that while day-to-day differences are modest for most of the period, wildfire significantly increases the magnitude of high-flow events, with many large peaks in the With-Fire simulation rising well above the 1:1 line. Together, the wavelet and distributional analyses demonstrate that wildfire has a statistically significant hydrologic impact, primarily due to intensification of storm-driven peak flows rather than a uniform change in all daily discharges.

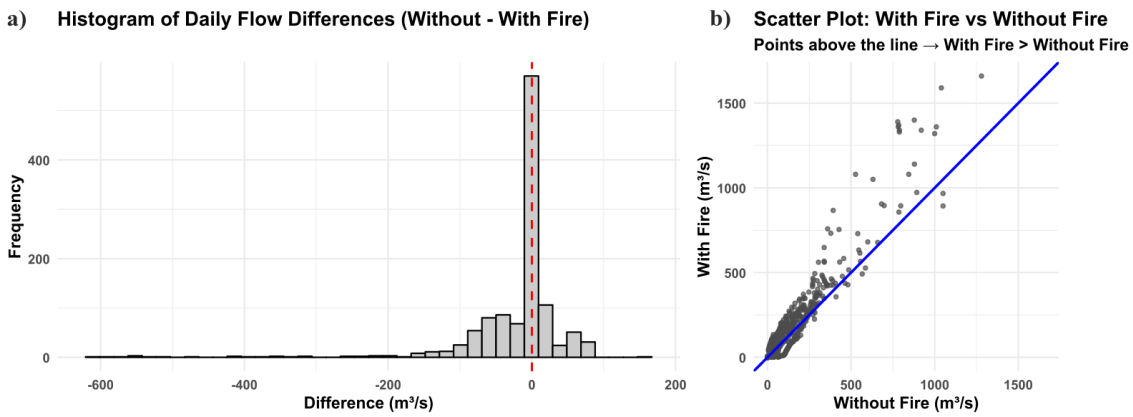


*Figure 6. A comparison of SWAT+ simulated streamflow with and without fire. Histogram (a), Scatter plots(b), show significantly increased post-fire flow, supported by Wilcoxon test result and Brunner test results.*

### 4.3 Monthly Hydrologic Response to Wildfire

The post-fire watershed exhibited a pronounced reorganization of hydrologic processes, with clear increases in surface runoff, lateral flow, and streamflow, mixed seasonal response in ET and strong reductions in deep percolation (**Figure 7**). These changes persisted throughout the 2021-2023 study period, indicating a sustained shift toward a faster, runoff-dominated system following wildfire.

Surface runoff showed substantial amplification after wildfire. Across all wet seasons (November-April), the With Fire simulation produced consistently higher runoff than the Without Fire baseline. For instance, monthly surface runoff nearly doubled in January 2023 (148 to 252 mm), and large increases were also observed in March 2023 (94.6 to 158 mm) and February 2021 (20.9 to 24.9 mm). Runoff in the dry season remained close to zero in both scenarios, reflecting rainfall limitation. The annual average spatial maps (2021 - 2023) (**Figure 8a**) confirm this pattern, with extensive positive differences concentrated in burned uplands, indicating that fire-induced reductions in infiltration resulted directly into greater overland flow

generation.

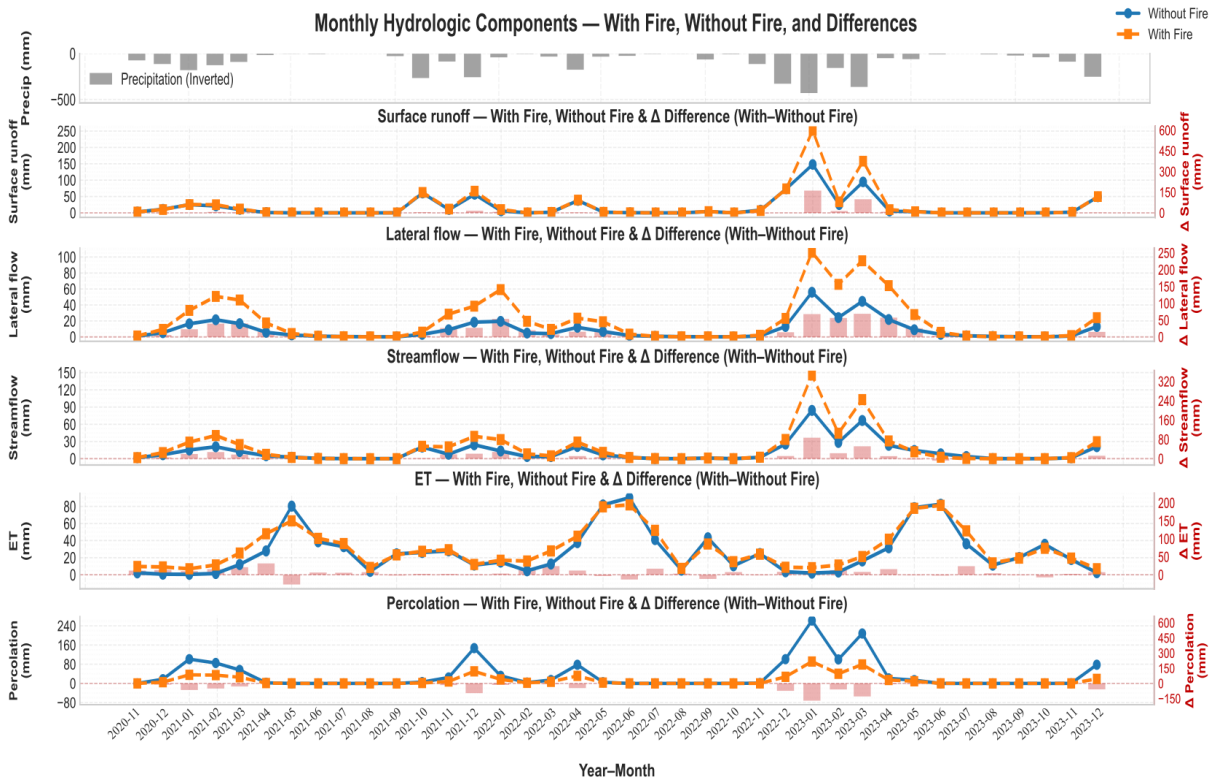


*Figure 7. Monthly hydrologic component dynamics under with fire and without-fire scenarios from 2020 November–2023 December. The panels show temporal variations in surface runoff, lateral flow, streamflow, ET, and percolation, along with precipitation (inverted bars) at the top. Blue solid lines represent simulations without fire, orange dashed lines indicate simulations with fire, and shaded red bars depict the difference (Δ) between with-fire and without-fire conditions. Secondary y-axes (right side) quantify these Δ values (With – Without Fire).*

Lateral flow, representing shallow subsurface runoff, also increased markedly under With Fire conditions. In high-rainfall months, lateral flow often doubled or tripled relative to the baseline (e.g., January 2021: 16.4 to 33.2 mm; January 2023: 55.9 to 106 mm). These increases reflect the formation of a near-surface hydrophobic layer that impedes vertical infiltration, causing water to accumulate in the upper soil layer and move downslope along shallow pathways. The average annual maps (**Figure 8b**) further extend this finding spatially: nearly all burned uplands have positive difference, reflecting a persistent deflection of subsurface flow along shallow pathways. The agreement between monthly maxima and spatially integrated average annual increments indicates that reduced percolation created hydrologic feedback-storing water in the near-surface zone and increasing horizontal redistribution downslope.

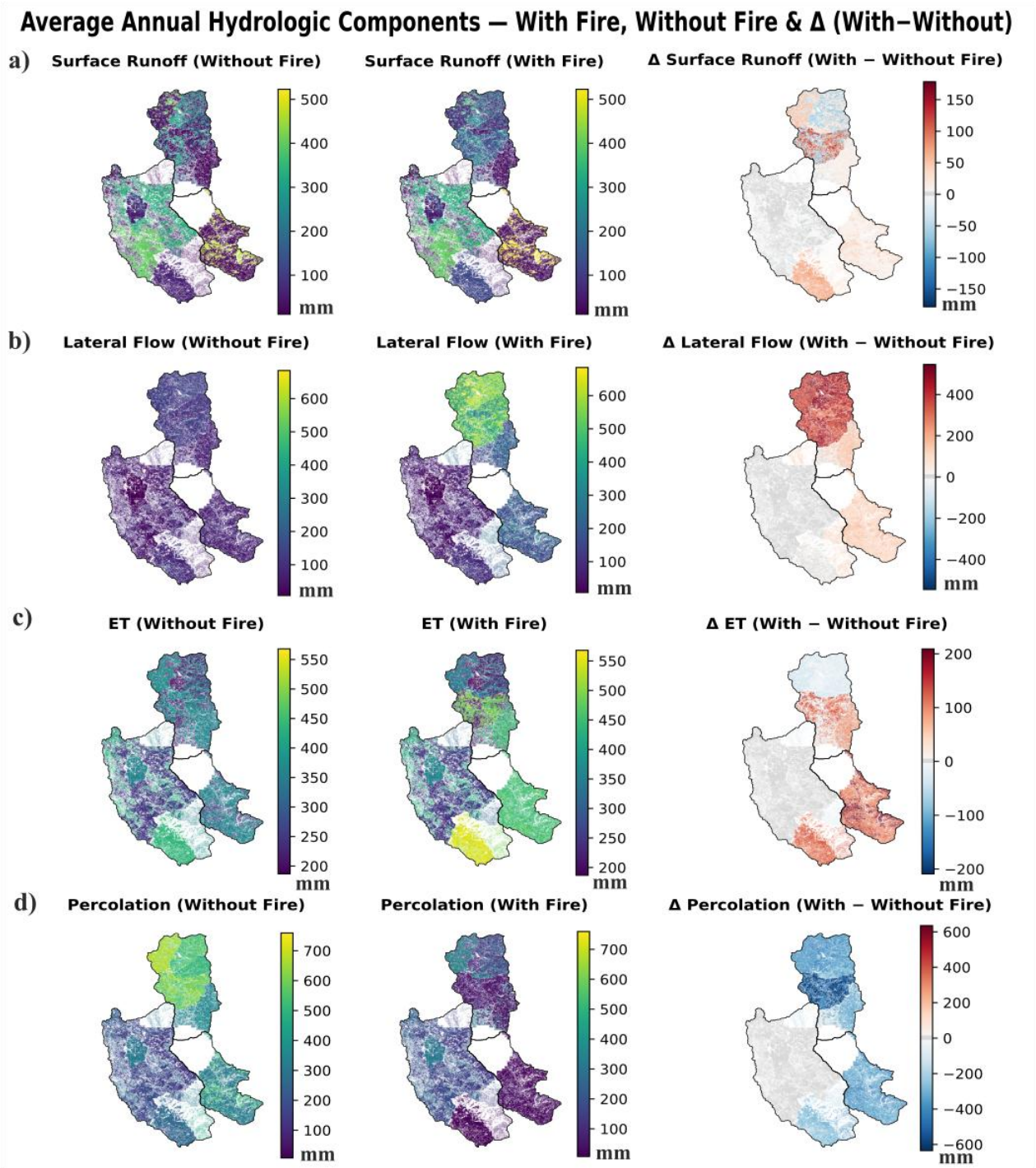


*Figure 8. Spatial distribution of average annual hydrologic components under with fire and without-fire conditions. Each row represents a hydrologic component - (a) Surface Runoff, (b) Lateral Flow, (c) ET, and (d) Percolation - averaged across 2021–2023. The first column shows spatial patterns simulated without fire, the second column shows patterns with fire, and the third column presents the difference (Δ = With − Without Fire).*

Streamflow responds to changes across runoff sources and shows a strong overall increase during wet periods. Major storm seasons produced large streamflow increment, such as in January 2021

(15.1 to 28.9 mm), January 2022 (13.4 to 33.3 mm), and January 2023 (84.2 to 146 mm). Throughout the wet season, the With Fire simulation consistently outperformed the Without Fire scenario, indicating enhanced responsiveness of the watershed. Small dry-season reductions occasionally occurred (e.g., August-September), likely due to diminished groundwater recharge resulting from reduced deep percolation earlier in the year. Overall, however, wildfire caused a major enhancement in event-scale and seasonal streamflow volume.ET displayed a seasonally mixed response, with clear directional shifts between dry and wet seasons. During warm, dry months, ET declined in With Fire conditions, likely due to canopy removal and reduced transpiration capacity. This is evident in May 2021 (80.3 to 63.1 mm) and May 2022 (78.5 to 77.8 mm). However, during cooler, wetter months, ET became consistently higher, driven by increased soil evaporation from exposed surfaces (e.g., April 2023: 31.6 to 41.6 mm). The average annual difference maps (**Figure 8c**) further display this cumulative effect: positive ΔET values define most of the burned area in those 3 years, indicating that the seasonal evaporative enhancement exceeded dry-season suppression. Spatially, the largest annual gains are apparent over burned catchment hillslopes where exposed soil and ash deposits maximized surface energy uptake and soil evaporation. Together, the monthly and average annual observations indicate a temporary, energy-driven boost to ET in the post-fire years, prior to vegetation re-establishment

Percolation experienced the most pronounced and persistent decline following wildfire. Reductions were visible during all wet months and reached severe magnitudes during high-rainfall events. For example, deep percolation decreased from 101 to 35.8 mm in January 2021, 147 to 50.3 mm in December 2021, and 262 to 91.1 mm in January 2023. These losses, often exceeding 50-200 mm/month, indicate a near-complete inhibition of vertical infiltration caused by hydrophobic soil layers and structural soil degradation. The average annual map (**Figure 8d**) shows that this deficit persists basin-wide throughout these three-year periods. The spatial distribution of negative percolation difference implies a basin-scale breakdown of deep infiltration and groundwater recharge, consistent with after-fire creation of soil hydrophobic layers and structural collapse. The agreement between temporal and spatial observations confirms that the infiltration barrier persisted over multiple rainy seasons, limiting vertical flow of water and reducing the basin's potential for long-term recharge.

The flow-to-precipitation (Q/P) ratio (**Figure 9**) further illustrates this hydrologic transformation. While pre-fire ratios (2007-2019) ranged from 0.30 to 0.65, the post-fire period (2021-2023) saw persistently high values, peaking at ~0.64 in 2023-even when annual precipitation remained similar to earlier years. Despite having similar annual precipitation, the flow-to-precipitation ratio is higher in the post-fire year 2022 exceeds the pre-fire years 2012 and 2015 (red dots), and 2023 surpasses 2011 and 2019 (blue dots). Between 2021 to 2023, the ratio shows a clear linear upward trend, culminating in a peak of ~0.64 in 2023, surpassed only by 2017-highlighting a steep growth trajectory that could continue into 2024 and beyond if current patterns persist. This persistent rise in flow-to-precipitation ratio underscores a lasting increase in runoff efficiency and loss of watershed buffering capacity.

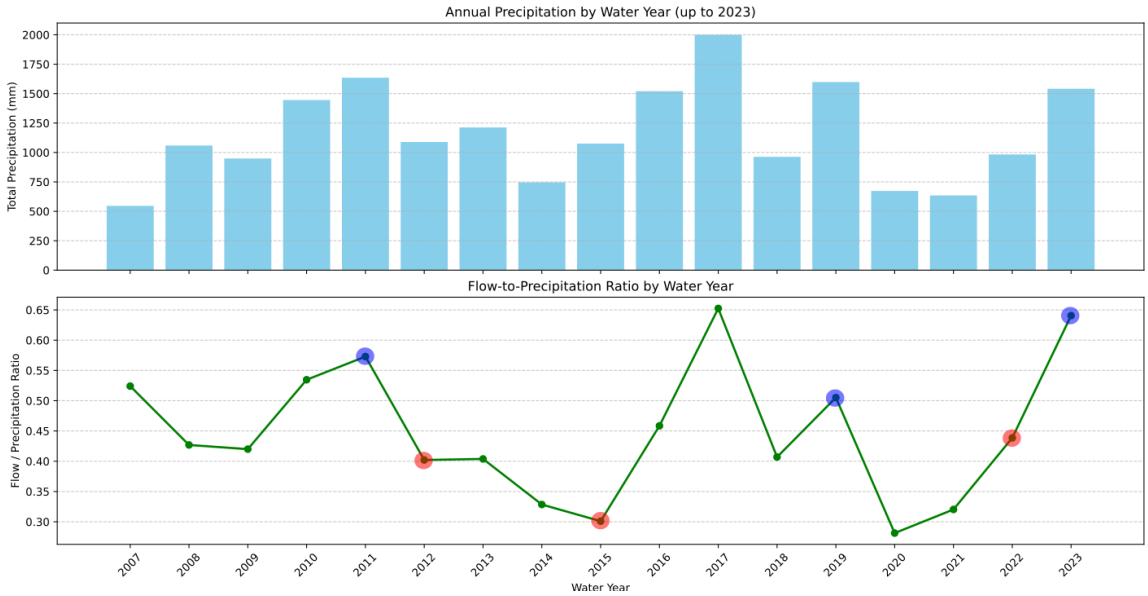


*Figure 9. Annual observed flow to precipitation ratio using USGS streamflow and PRISM precipitation data*

Collectively, the post-fire hydrologic system is governed by a cascading set of relationships. The variation in ET alters overall water availability in the system. Fire-affected soils with reduced infiltration capacity ultimately led to infiltration failure. This failure triggers two key flow-path shifts: first, surface runoff rises, increasing streamflow peaks; second, the percolation deficit redirects water into lateral flow, further enhancing rapid streamflow generation. Together, these changes represent a fundamental transition from slow, vertical, and consumptive pathways (ET and Percolation) to fast, horizontal export pathways (Surface runoff and lateral flow).

# 5. Discussions

## 5.1 Model Performance and Wildfire Impacts on Streamflow

Hydrologic modeling in post-wildfire settings is often difficult because predictions are very uncertain and many different parameter sets can produce similar streamflow even when they do not represent the true internal processes of the catchment (Beven, 2001; Ebel et al., 2023). This study addressed that problem with a dual calibration strategy. A key result is that the Data-Driven (Bidirectional) calibration clearly outperformed the Literature-Constrained (Unidirectional) approach. The Literature-Constrained model achieved a reasonable Nash-Sutcliffe Efficiency (NSE = 0.582), but its P-factor was only 11.0%, meaning that the uncertainty band failed to include almost 90% of the observed flow variability. This shows that strict use of standard rules of thumb (e.g., forcing curve numbers only upward or canopy storage only downward) can be too rigid and may hide important post-fire response, such as the trade-off between ash water storage and surface sealing (Woods & Balfour, 2010).

In contrast, the Bidirectional model allowed parameters to move up or down within symmetric bounds and achieved a P-factor of 78.9% and an NSE of 0.678. By naturally converging toward

physically reasonable values, mainly lower hydraulic conductivity and higher soil evaporation compensation, without being forced at the start, the model provided an independent check on the strength of the wildfire signal. Based on the guidelines of Abbaspour et al. (2018) for disturbed watersheds, the ability of the bidirectional model to capture the uncertainty bands makes it the more reliable option for risk assessment.

The statistical strength of this hydrologic change was also tested with non-parametric methods. The Wilcoxon signed-rank test ($p < 0.05$) and the Brunner-Munzel test ($p < 0.05$) both rejected the null hypothesis that the distributions are the same, showing that the post-fire flow regime is statistically different from the pre-fire baseline. Wavelet analysis (Level 4 approximation) further showed a strong increase in low-frequency power during the wet seasons. Together, these results indicate that wildfire did not just add random noise to the hydrograph, it also changed how the watershed converts precipitation into streamflow, shifting the system from a buffered, resistance-dominated regime to a more responsive, flashy regime (Hallema et al., 2017).

### 5.2 Seasonal Controls and Interdependent Changes of Hydrologic Variables

The post-fire hydrologic regime of the Middle Fork Eel Watershed was not a simple, linear reaction to vegetation loss. Instead, the model showed a complex shift in the watershed's dynamics. This reorganization was driven by a fundamental breakdown in vertical soil connectivity, where the interaction of specific parameter changes produced a new hydrophobic ash-seal structure.

The primary driver of the hydrologic shift was a strong restriction of vertical connectivity. In Moderate Severity zones, the calibrated Saturated Hydraulic Conductivity (K) fell by up to 50%, while the Soil Evaporation Compensation Factor (ESCO) increased at the same time to its upper limit (+50%). This pairing of parameters points to a key physical mechanism: fire-induced hydrophobicity clogged soil macropores and shut down deep percolation, while the ash layer on top acted as a mineral mulch. As described by Stoof et al. (2010) and Woods and Balfour (2010), this creates a hydraulic paradox where the surface becomes almost impermeable to infiltration, which produces more runoff, while deeper soil moisture is cut off from atmospheric demand, which stops capillary rise and evaporation.

This sealing of the vertical pathway forced a shift in flow directions from vertical to horizontal. The model showed a large drop in percolation, on the order of 50-200 mm/month during peak wet seasons (Figure 7). Because this water could no longer move downward to recharge the deep aquifer, it was redirected into lateral flow. The strong increase in lateral flow suggests that even when water passed through the top ash layer, it soon met a patchy hydrophobic zone at shallow depths and was pushed to travel downslope through the upper soil instead of percolating deeper. This supports the subsurface bypass mechanism described by Nyman et al. (2010), in which macropore flow is activated in water-repellent soils and quickly delivers water to channels without interacting with deeper storage.

At the same time, the watershed lost much of its ability to store water in the soil. Available Water Capacity (AWC) decreased by about 22-26% across all burn-severity classes. This drop indicates that combustion of Soil Organic Matter (SOM) destroyed soil aggregates that hold water, effectively removing the sponge function of the watershed (Ebel et al., 2012). Soil Bulk Density (BD) decreased by 25-44%, which reflects the mixing of low-density ash into the mineral soil. Usually, a lower bulk density would suggest higher porosity, but the combined drop in AWC shows that this extra porosity was structural (large, poorly connected voids) rather than matric (small, water-holding pores), which speeds drainage instead of storing water.

Evapotranspiration (ET) showed a seasonally split response that does not match simple vegetation loss explanations. During the dry season (May-October), ET decreased because the canopy that once transpired water (CANMX) was destroyed and because the hydrophobic seal limited extraction of water. During the wet season (November-April), ET actually increased. This pattern can be explained by the Aerodynamic Exposure concept of Meili et al. (2023). After the fire, canopy removal increased wind speed and turbulence at the surface. Combined with high near-surface soil moisture maintained by the ash mulch (high ESCO), this aerodynamic forcing produced strong physical soil evaporation during winter periods, which at times exceeded the lost transpiration. This agrees with the findings of Baur et al. (2024), who showed that fire speeds up soil moisture loss mainly through physical pathways and weakens the usual link between plants and soil water.

The combined effect of these changes-infiltration failure, lateral rerouting, loss of storage, and aerodynamic exposure-turned the watershed into a tipping bucket system. This finding is further supported by the Flow-to-Precipitation (Q/P) ratio, which increased in a nearly linear way and reached about 0.64 in 2023 (**Figure 9**). This sustained rise shows that the watershed has largely lost its delay between rainfall and runoff: precipitation is now turned into effective discharge much more quickly. This flashiness was further strengthened by a reduction in Overland Flow Manning's Roughness (OVN), which smoothed hillslopes and increased runoff speed, producing the high peak hydrographs that define the post-fire regime (Hallema et al., 2017).

### 5.3 Implications for Watershed Management and Global Significance

The hydrologic changes found in the Middle Fork Eel Watershed act as a warning for other fire-prone Mediterranean-type regions. The move from a resistance-dominated system, with deep storage and slow release, to a flashy, export-dominated system has serious consequences for water security in a time of expanding megadroughts (Williams et al., 2022).

This study supports the worrying global pattern described by Baur et al. (2024). Wildfire acts as a hydrological accelerator that speeds up aridification. By turning rainfall quickly into surface runoff, the watershed loses the chance to recharge deep soil and groundwater. At a global scale, this means that burned regions from Australia to southern Europe may face a flashy but dry paradox. They will see more frequent and intense floods during storms, partly due to changes in CN2 and OVN, but they will also suffer stronger hydrologic droughts between storms because

percolation has collapsed. This process directly threatens the resilience of dryland ecosystems that rely on deep soil water stored from the wet season.

The stronger peak flows seen here also demand a re-evaluation of hydraulic infrastructure design. The flashier post-fire regime implies that the Time of Concentration (Tc) of watersheds is now shorter than it was in the past. As a result, short-duration, high-intensity convective storms, which are becoming more common with climate change, can cause severe floods in burned basins that were previously buffered by forest litter and deep soil storage. This creates an urgent need to update Intensity-Duration-Frequency (IDF) curves used to design culverts, bridges, and reservoirs in fire-prone regions around the world (Hallema et al., 2017).

Moreover, the long-lasting decline in deep percolation (**Figure 8d**) also threatens groundwater sustainability. In many Mediterranean climates, mountain watersheds act as water towers that capture winter precipitation and recharge alluvial aquifers. The results of this study suggest that wildfire can effectively replace these recharge areas with a hydrophobic seal. If this mechanism is extended to the growing burned areas in the western United States and Chile, it implies a long-term drop in aquifer replenishment that is not yet included in global groundwater models (Ebel et al., 2023). Therefore, post-fire management needs to go beyond erosion control and include treatments that physically break up hydrophobic layers and restore vertical infiltration pathways.

### 5.4 Limitations and Recommendations

Although the SWAT+ model successfully reproduced the main hydrologic shifts, this study still has limits that come from using a semi-distributed model. The HRU-based setup assumes that each land-use and soil combination is uniform, which may oversimplify how hydrophobic patches connect across hillslopes (Famiglietti & Wood, 1995). In addition, although the bidirectional calibration gave strong indirect evidence of parameter changes, there were no direct field measurements of soil hydrophobicity or ash thickness to confirm the exact K and ESCO values. Future research should focus on field validation by carrying out in-situ hydraulic conductivity tests to check the size of the ash seal effect estimated by the model on spatial discretization by using grid-based or fully distributed routing schemes to better capture how burned hillslopes connect to the channel network, and on long-term recovery by extending the simulation period to find how long it takes for the watershed to return toward pre-fire conditions.

## 6. Conclusion

In conclusion, this study shows a new and more reliable way to model post-wildfire hydrology by using a data-driven, bidirectional calibration framework instead of traditional literature-constrained methods. When parameters were allowed to move freely in both directions, the model naturally settled into a clearly different post-fire state and reached high predictive reliability, something the rigid, one-directional parameter changes could not match. This success gives confidence that the hydrologic changes we see are not just side effects of the model setup, but a real signal of a major shift in how the watershed behaves.

In physical terms, the August Complex Wildfire did more than remove vegetation: it altered the vertical link in the soil profile by creating a hydrophobic ash seal. The combination of reduced hydraulic conductivity and higher soil evaporation compensation formed a cover on the soil that shut down deep percolation and groundwater recharge, forcing water to move from vertical storage to horizontal export. The watershed lost its sponge behavior and became a flashy, export-dominated system, as shown by the steady rise in the surface runoff and lateral flow and decline in percolation. At a broader scale, these results highlight a flashy but dry paradox for fire-prone regions: after fire, watersheds may send more water to streams during storms (a tipping bucket effect), but this comes at the cost of the deep storage needed to support baseflows during dry periods. As climate change increases the size and severity of burned areas, this temporary capping of recharge zones becomes a quiet but serious threat to long-term water security. This means future watershed management must go beyond simple erosion control and actively work to restore vertical infiltration pathways so that headwater basins can keep functioning as reliable water supply source in a warming world.

## Data Availability Statement

The SWAT+ public version can be downloaded from (https://swat.tamu.edu/software/plus/) and the source code capable of simulating LULC change can be obtained from (https://github.com/swat-model/swatplus/tree/main/). Streamflow data can be accessed from USGS National Water Dashboard (https://dashboard.waterdata.usgs.gov/app/nwd/en/). All the data sets used are available and can be accessed publicly. The SWAT+ model simulations with input variables from this study are available for download at [*link to be added from Zenodo, currently being setup with an ODC Attribution (ODC-BY) license for access without restrictions*].

## Acknowledgements

This research is generously funded by the U.S. Department of Energy under Grant No. DE-EM0005314.

## References

Abbaspour, K. C., Johnson, C. A., & van Genuchten, M. T. (2004). Estimating Uncertain Flow and Transport Parameters Using a Sequential Uncertainty Fitting Procedure. Vadose Zone Journal, 3(4), 1340–1352. https://doi.org/10.2136/vzj2004.1340

Abbaspour, K. C., Vaghefi, S. A., & Srinivasan, R. (2018). A Guideline for Successful Calibration and Uncertainty Analysis for Soil and Water Assessment: A Review of Papers from the 2016 International SWAT Conference. *Water*, *10*(1), 6. https://doi.org/10.3390/w10010006

Aldrich, E. (2020). *wavelets: Functions for computing wavelet filters, wavelet transforms and multiresolution analyses* (R package version 0.3-0.2). Retrieved from https://cran.r-project.org/web/packages/wavelets/wavelets.pdf (accessed 20 February 2025).

Arnold, J. G., Srinivasan, R., Muttiah, R. S., & Williams, J. R. (1998). Large area hydrologic modeling and assessment Part I: Model development. *JAWRA Journal of the American Water Resources Association, 34*(1), 73–89. https://doi.org/10.1111/j.1752-1688.1998.tb05961.x

Arnold, J. G., Čerkasova, N., White, M. J., Bailey, R., Thorp, K., Jeong, J., Zhang, X., Ugraskan, T., van Griensven, A., Rathjens, H., Raj, C., Cai, X., Geter, W. F., David, O., Carlson, J. R., & Le, K. N. (2024). Soil and Water Assessment Tool Plus (SWAT+) [Computer software]. https://github.com/swat-model/swatplus

Batelis, SC., Nalbantis, I. (2014). Potential Effects of Forest Fires on Streamflow in the Enipeas River Basin, Thessaly, Greece. Environ. Process. 1, 73–85 https://doi.org/10.1007/s40710-014-0004-z

Baur, M.J., Friend, A.D. & Pellegrini, A.F.A. (2024). Widespread and systematic effects of fire on plant–soil water relations. *Nat. Geosci.* **17**, 1115–1120. https://doi.org/10.1038/s41561-024-01563-6

Benavides-Solorio, J., & MacDonald, L. H. (2001). Post-fire runoff and erosion from simulated rainfall on small plots, Colorado Front Range. *Hydrological Processes, 15*, 2931–2952. https://doi.org/10.1002/hyp.383

Beven, K. (2001). How far can we go in distributed hydrological modelling? *Hydrology and Earth System Sciences, 5*(1), 1–12. https://doi.org/10.5194/hess-5-1-2001

Bieger, K., Arnold, J. G., Rathjens, H., White, M. J., Bosch, D., Allen, P. M., Volk, M., & Srinivasan, R. (2017). Introduction to SWAT+, a completely restructured version of the Soil and Water Assessment Tool. *Journal of the American Water Resources Association, 53*(1), 115–130. https://doi.org/10.1111/1752-1688.12482

Bihon, Y. T., Lohani, T. K., Ayalew, A. T., Neka, B. G., Mohammed, A. K., Geremew, G. B., & Ayele, E. G. (2024). Performance evaluation of various hydrological models with respect to hydrological responses under climate change scenario: a review. *Cogent Engineering*, *11*(1). https://doi.org/10.1080/23311916.2024.2360007

Bladon, K. D., Emelko, M. B., Silins, U., & Stone, M. (2014). Wildfire and the future of water supply. *Environmental Science & Technology, 48*, 8936–8943. https://doi.org/10.1021/es500130g

Brunner, E., & Munzel, U. (2000). The nonparametric Behrens-Fisher problem: Asymptotic theory and a small-sample approximation. Biometrical Journal, 42(1), 17–25. https://doi.org/10.1002/(SICI)1521-4036(200001)42:1<17::AID-BIMJ17>3.0.CO;2-U

CAL FIRE (2023). August Complex (includes Doe Fire). https://www.fire.ca.gov/incidents/2020/8/16/august-complex-includes-doe-fire/ (accessed 10 February 2025).

Celray James CHAWANDA. (2022). SWAT+ Toolbox (0.7.6). Zenodo. https://doi.org/10.5281/zenodo.6331716

Clark, M. P., Bierkens, M. F. P., Samaniego, L., Woods, R. A., Uijlenhoet, R., Bennett, K. E., Pauwels, V. R. N., Cai, X., Wood, A. W., & Peters-Lidard, C. D. (2017). The evolution of process-based hydrologic models: Historical challenges and the collective quest for physical realism. *Hydrology and Earth System Sciences, 21*, 3427–3440. https://doi.org/10.5194/hess-21-3427-2017

Coombs, J. S., & Melack, J. M. (2013). Initial impacts of a wildfire on hydrology and suspended sediment and nutrient export in California chaparral watersheds. *Hydrol. Process., 27*, 3842–3851. https://doi.org/10.1002/hyp.9508

Daly, C., Halbleib, M., Smith, J.I., Gibson, W.P., Doggett, M.K., Taylor, G.H., Curtis, J. and Pasteris, P.P. (2008), Physiographically sensitive mapping of climatological temperature and precipitation across the conterminous United States. Int. J. Climatol., 28: 2031-2064. https://doi.org/10.1002/joc.1688

Doerr, S. H., & Santín, C. (2016). Global trends in wildfire and its impacts: Perceptions versus realities in a changing world. *Philosophical Transactions of the Royal Society B: Biological Sciences, 371*, 20150345. https://doi.org/10.1098/rstb.2015.0345

Ebel, B. A. (2012), Wildfire impacts on soil-water retention in the Colorado Front Range, United States, *Water Resour. Res.*, 48, W12515, doi:10.1029/2012WR012362.

Ebel, B. A., & Moody, J. A. (2020). Parameter estimation for multiple post-wildfire hydrologic models. *Hydrological Processes, 34*, 4049–4066. https://doi.org/10.1002/hyp.13865

Ebel, B. A., Shephard, Z. M., Walvoord, M. A., Murphy, S. F., Partridge, T. F., & Perkins, K. S. (2023). Modeling post-wildfire hydrologic response: Review and future directions for applications of physically based distributed simulation. *Earth's Future, 11*(2), e2022EF003338. https://doi.org/10.1029/2022EF003038

Eidenshink, J., Schwind, B., Brewer, K., Zhu, Z., Quayle, B., & Howard, S. (2007). A project for monitoring trends in burn severity. *Fire Ecology, 3*(1), 3–21. https://doi.org/10.4996/fireecology.0301003

Esmaeilzadeh, M., Amirzadeh, M. (2024). Replication Study: Enhancing Hydrological Modeling with Physics-Guided Machine Learning. https://doi.org/10.48550/arXiv.2402.13911

Famiglietti, J. S., & Wood, E. F. (1995). Effects of spatial variability and scale on areally averaged evapotranspiration. *Water Resources Research, 31*, 699–712. https://doi.org/10.1029/94WR02820

FAO/UNESCO (1974–1981). Soil Map of the World (1:5,000,000 scale), 10 volumes. Food and Agriculture Organization of the United Nations and UNESCO, Paris. Available at: https://data.apps.fao.org/map/catalog/srv/eng/catalog.search#/metadata/cc45a270-88fd-11da-a88f-000d939bc5d8 (accessed 10 February 2025).

Farr, T. G., et al. (2007), The Shuttle Radar Topography Mission, *Rev. Geophys.*, 45, RG2004, doi:10.1029/2005RG000183.

Gaucherel, C. (2002). Use of wavelet transform for temporal characterisation of remote watersheds. *Journal of Hydrology, 269*(3–4), 101–121. https://doi.org/10.1016/S0022-1694(02)00212-3

Gupta, H. V., Kling, H., Yilmaz, K. K., and Martinez, G. F. (2009). *Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. Journal of Hydrology, 377,* 80–91. https://doi.org/10.1016/j.jhydrol.2009.08.003

Hallema, D. W., Sun, G., Caldwell, P. V., Norman, S. P., Cohen, E. C., Liu, Y., Ward, E. J., & McNulty, S. G. (2018). Burned forests impact water supplies. *Nature Communications, 9*(1), 1307. https://doi.org/10.1038/s41467-018-03735-6

Hallema, D. W., Sun, G., Bladon, K. D., Norman, S. P., Caldwell, P. V., Liu, Y., & McNulty, S. G. (2017). Regional patterns of postwildfire streamflow response in the Western United States: The importance of scale-specific connectivity. *Hydrological Processes, 31*, 2582–2598. https://doi.org/10.1002/hyp.11208

Havel, A., Tasdighi, A., & Arabi, M. (2018). Assessing the hydrologic response to wildfires in mountainous regions. *Hydrology and Earth System Sciences, 22*, 2527–2550. https://doi.org/10.5194/hess-22-2527-20183

Inbar, M., Tamir, M., & Wittenberg, L. (1998). Runoff and erosion processes after a forest fire in Mount Carmel, a Mediterranean area. *Geomorphology, 24*, 17–33. https://doi.org/10.1016/S0169-555X(97)00098-6

J.A.H. Brown. (1972), Hydrologic effects of a bushfire in a catchment in south-eastern New South Wales, *Journal of Hydrology*, 15(1), 77-96, ISSN 0022-1694. https://doi.org/10.1016/0022-1694(72)90077-7

Jansen LS, O'Dowd A, Bouma-Gregson K. (2020). A comparison of benthic algal and macroinvertebrate communities in a dammed and undammed Mediterranean river (Eel River watershed, California, USA). *River Res Applic*. 36, 1668–1681. https://doi.org/10.1002/rra.3695

Kiesel, J., Schmalz, B., Brown, G. L., & Fohrer, N. (2013). Application of a hydrological-hydraulic modelling cascade in lowlands for investigating water and sediment fluxes in catchment, channel and reach. *Journal of Hydrology and Hydromechanics, 61*, 334–346. https://doi.org/10.2478/johh-2013-0042

Knoben, W. J. M., Freer, J. E., and Woods, R. A. (2019). Inherent benchmark or not? Comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. *Hydrology and Earth System Sciences*, 23, 4323–4331. https://doi.org/10.5194/hess-23-4323-2019

Labat, D., Ouarda, T. B. M. J., Travassos e Silva, F. F., & Rodrigues, I. (2000). Rainfall-runoff relations for karstic springs. Part II: Continuous wavelet and discrete orthogonal multiresolutions analyses. *Journal of Hydrology, 238*(3–4), 149-178. https://doi.org/10.1016/S0022-1694(00)00322-X.

Liu, S., Leslie, L. M., Speer, M., Bunker, R., & Mo, X. (2004). The effects of bushfires on hydrological processes using a paired-catchment analysis. *Meteorology and Atmospheric Physics, 86*, 31–44. https://doi.org/10.1007/s00703-003-0614-x

Loáiciga, H. A., Pedreros, D., & Roberts, D. (2001). Wildfire-streamflow interactions in a chaparral watershed. *Advances in Environmental Research, 5(3), 295–305. https://doi.org/10.1016/S1093-0191(00)00064-2*

Ma, Q., Bales, R. C., Rungee, J., Conklin, M. H., Collins, B. M., & Goulden, M. L. (2020). Wildfire controls on evapotranspiration in California's Sierra Nevada. *Journal of Hydrology, 590,* 125364. https://doi.org/10.1016/j.jhydrol.2020.125364

Mahat, V., Silins, U., & Anderson, A. (2016). Effects of wildfire on the catchment hydrology in southwest Alberta. *CATENA, 147*, 51–60. https://doi.org/10.1016/j.catena.2016.06.040

Maina, F. Z., & Siirila-Woodburn, E. R. (2020). Watersheds dynamics following wildfires: Nonlinear feedbacks and implications on hydrologic responses. *Hydrological Processes, 34*, 33–50. https://doi.org/10.1002/hyp.13568

Martin, D. A., & Moody, J. A. (2001). Comparison of soil infiltration rates in burned and unburned mountainous watersheds. *Hydrological Processes, 15*, 2893–2903. https://doi.org/10.1002/hyp.380

McDonald, S., Mohammed, I.N., Bolten, J.D., Pulla, S., Meechaiya, C., Markert, A., Nelson, E.J., Srinivasan, R., Lakshmi, V. (2019). Web-based decision support system tools: The Soil and Water Assessment Tool Online visualization and analyses (SWATOnline) and NASA earth observation data downloading and reformatting tool (NASAaccess). *Environmental Modelling & Software*, 120, 104499. https://doi.org/10.1016/j.envsoft.2019.104499

Meili, N., Beringer, J., Zhao, J., & Fatichi, S. (2023). Aerodynamic effects cause higher forest evapotranspiration and water yield reductions after wildfires in tall forests. *Global Change Biology, 29*(24), 6918–6935. https://doi.org/10.1111/gcb.16995

Moody, J. A., & Martin, D. A. (2001). Initial hydrologic and geomorphic response following a wildfire in the Colorado front range. *Earth Surface Processes and Landforms, 26*(10), 1049–1070. https://doi.org/10.1002/esp.253

Moody, J. A., Shakesby, R. A., Robichaud, P. R., Cannon, S. H., & Martin, D. A. (2013). Current research issues related to post-wildfire runoff and erosion processes. *Earth-Science Reviews, 122*, 10–37. https://doi.org/10.1016/j.earscirev.2013.03.004

Murphy, S. F., McCleskey, R. B., Martin, D. A., Writer, J. H., & Ebel, B. A. (2018). Fire, flood, and drought: Extreme climate events alter flow paths and stream chemistry. *Journal of Geophysical Research: Biogeosciences, 123*(8), 2513–2526. https://doi.org/10.1029/2017JG004349

Nasirzadehdizaji, R., & Akyuz, D. E. (2022). Predicting the potential impact of forest fires on runoff and sediment loads using a distributed hydrological model. *Ecological Modelling, 468*, 109959. https://doi.org/10.1016/j.ecolmodel.2022.109959

Neary, D. G., Gottfried, G. J., & Ffolliott, P. F. (2003). Post-wildfire watershed flood responses. In *Paper 1B7, Second International Fire Ecology and Fire Management Congress, 16–20 November 2003, Orlando, Florida*.

Neitsch, S. L., Arnold, J. G., Kiniry, J. R., & Williams, J. R. (2011). *Soil and water assessment tool theoretical documentation version 2009* (Texas Water Resources Institute Technical Report No. 406, p. 618).

Niroula, Sundar, Kevin Wallington and Ximing Cai. (2023). Addressing data challenges in riverine nutrient load modeling of an intensively managed agro-industrial watershed. *Journal of the American Water Resources Association,* 59(2), 213–225. https://doi.org/10.1111/1752-1688.13097

Nkwasa, A., Chawanda, C.J., van Griensven, A., 2022. Regionalization of the SWAT+ model for projecting climate change impacts on sediment yield: An application in the Nile basin. *Journal of Hydrology: Regional Studies*, 42, 101152. https://doi.org/10.1016/j.ejrh.2022.101152

Nyman, P., Sheridan, G. and Lane, P.N.J. (2010), Synergistic effects of water repellency and macropore flow on the hydraulic conductivity of a burned forest soil, south-east Australia. Hydrol. Process., 24: 2871-2887. https://doi.org/10.1002/hyp.7701

Pierson, F.B., Robichaud, P.R. and Spaeth, K.E. (2001), Spatial and temporal effects of wildfire on the hydrology of a steep rangeland watershed[†]. Hydrol. Process., 15: 2905-2916. https://doi.org/10.1002/hyp.381

PRISM Group (2014). Parameter-elevation Regressions on Independent Slopes Model (PRISM) Climate Data. Oregon State University, Corvallis, OR. https://prism.oregonstate.edu/ (accessed on 15 February 2025).

R Documentation Contributors (2023). *wilcox.test: Wilcoxon Rank Sum and Signed Rank Tests* (Stats package version 3.6.2). RDocumentation. https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/wilcox.test (accessed 20 February 2025).

Robichaud, P. R. (2000). Fire effects on infiltration rates after prescribed fire in Northern Rocky Mountain forests, USA. *Journal of Hydrology, 231–232*, 220–229. https://doi.org/10.1016/S0022-1694(00)00196-7

Robinne, F.-N., Hallema, D. W., Bladon, K. D., et al. (2021). Scientists' warning on extreme wildfire risks to water supply. *Hydrological Processes, 35*(9), e14086. https://doi.org/10.1002/hyp.14086

Ryan, S. E., Dwire, K. A., & Dixon, M. K. (2011). Impacts of wildfire on runoff and sediment loads at Little Granite Creek, western Wyoming. *Geomorphology, 129*, 113–130. https://doi.org/10.1016/j.geomorph.2011.01.017

Saunders, W. C., N. Bouwes, P. McHugh, and C. E. Jordan. (2018). A network model for primary production highlights linkages between salmonid populations and autochthonous resources. *Ecosphere* 9(3), e02131. 10.1002/ecs2.2131

Schaefli, B., Maraun, D., & Holschneider, M. (2007). What drives high flow events in the Swiss Alps? Recent developments in wavelet spectral analysis and their application to hydrology. *Advances in Water Resources, 30*(12), 2511–2525. https://doi.org/10.1016/j.advwatres.2007.06.004

Scott, D. F., & Van Wyk, D. B. (1990). The effects of wildfire on soil wettability and hydrological behaviour of an afforested catchment. *Journal of Hydrology, 121*, 239–256. https://doi.org/10.1016/0022-1694(90)90234-O

Scott, D. N. and Wohl, E. E. (2018). Geomorphic regulation of floodplain soil organic carbon concentration in watersheds of the Rocky and Cascade Mountains, USA. Earth Surf. Dynam., 6, 1101–1114, https://doi.org/10.5194/esurf-6-1101-2018.

Shakesby, R. A., & Doerr, S. H. (2006). Wildfire as a hydrological and geomorphological agent. *Earth-Science Reviews, 74*, 269–307. https://doi.org/10.1016/j.earscirev.2005.10.006

Smith, H. G., Sheridan, G. J., Lane, P. N. J., Nyman, P., & Haydon, S. (2011). Wildfire effects on water quality in forest catchments: A review with implications for water supply. *Journal of Hydrology, 396*, 170–192. https://doi.org/10.1016/j.jhydrol.2010.10.043

Stoof, C. R., Wesseling, J. G., & Ritsema, C. J. (2010). Effects of fire and ash on soil water retention. *Geoderma, 159*(3–4), 276–285. https://doi.org/10.1016/j.geoderma.2010.08.002

SWAT+ Development Team (2025). *SWAT+ Software*. https://swat.tamu.edu/software/plus (accessed 1 February 2025).

Torrence, C., & Compo, G. P. (1998). A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society, 79*(1), 61–78. https://doi.org/10.1175/1520-0477(1998)079%3C0061:APGTWA%3E2.0.CO;2

Toshiaki, A. (2022). brunnermunzel: Brunner-Munzel Test. R package version 2.0. https://cran.r-project.org/web/packages/brunnermunzel/brunnermunzel.pdf

Townsend, S. A., & Douglas, M. M. (2004). The effect of a wildfire on stream water quality and catchment water yield in a tropical savanna excluded from fire for 10 years (Kakadu National Park, North Australia). *Water Research, 38*, 3051–3058. https://doi.org/10.1016/j.watres.2004.04.009

U.S. Geological Survey (USGS) (2024), Annual NLCD Collection 1 Science Products (ver. 1.1, June 2025): U.S. Geological Survey data release, https://doi.org/10.5066/P94UXNTS. (accessed 10 February 2025).

United States Geological Survey (USGS) & U.S. Forest Service (2025). Monitoring Trends in Burn Severity (MTBS) program – Burn Severity Viewer. https://burnseverity.cr.usgs.gov/viewer/?product=MTBS (accessed 15 February 2025).

United States Geological Survey (USGS) (2018). *USGS EROS Archive – Digital Elevation – Shuttle Radar Topography Mission (SRTM) 1 Arc-Second Global*. https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm-1 (accessed 1 February 2025). DOI:10.5066/F7PR7TFT

United States Geological Survey (USGS) (2025a). National Water Dashboard. U.S. Geological Survey. https://dashboard.waterdata.usgs.gov/app/nwd/en/ (accessed 12 February 2025).

United States Geological Survey (USGS) (2025b). Post-Fire Mapping Glossary – Burn Severity Portal. https://burnseverity.cr.usgs.gov/glossary (accessed 20 February 2025).

Venkatesh, K., Preethi, K., & Ramesh, H. (2020). Evaluating the effects of forest fire on water balance using fire susceptibility maps. *Ecological Indicators, 110*, 105856. https://doi.org/10.1016/j.ecolind.2019.105856

Wampler, K. A., Bladon, K. D., & Faramarzi, M. (2023). Modeling wildfire effects on streamflow in the Cascade Mountains, Oregon, USA. *Journal of Hydrology, 621*, 129585. https://doi.org/10.1016/j.jhydrol.2023.129585

Wilcoxon, F. (1945). Individual Comparisons by Ranking Methods. Biometrics Bull. 1, 80–83. https://doi.org/10.2307/3001968

Williams, A. P., Livneh, B., McKinnon, K. A., Hansen, W. D., Mankin, J. S., Cook, B. I., et al. (2022). Growing impact of wildfire on Western US water supply. *Proceedings of the National Academy of Sciences, 119*(10), e2114069119. https://doi.org/10.1073/pnas.2114069119

Wine, M. L., & Cadol, D. (2016). Hydrologic effects of large southwestern USA wildfires significantly increase regional water supply: Fact or fiction? *Environmental Research Letters, 11*, 085006. https://doi.org/10.1088/1748-9326/11/8/085006

Woods, S. W., & Balfour, V. N. (2010). The effects of soil texture and ash thickness on the post-fire hydrological response from ash-covered soils. Journal of Hydrology, 393(3–4), 274–286. https://doi.org/10.1016/j.jhydrol.2010.08.025